Siyu Liang

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Research interests

My research interests are phonetics, computational linguistics, field work, as well as historical linguistics and typology. I am particularly interested in low-resource speech NLP, language technology, and quantitative modeling of speech.

Education

2021 – Present **University of Washington** – Seattle, WA
Ph.D. in Computational Linguistics
Advisors: Professors Gina-Anne Levow & Richard Wright

2025 **University of Washington** – Seattle, WA
M.S. in Computational Linguistics

2020 **Georgetown University** – Washington, DC
B.A. in Linguistics with minors in Persian & Computer Science
Advisor: Professor Elizabeth Zsiga

Publications

Siyu Liang, Gina-Anne Levow. 2025. Breaking the Transcription Bottleneck: Fine-tuning ASR Models for Extremely Low-Resource Fieldwork Languages . In *the 4th Workshop on NLP Applications to Field Linguistics (FieldMatters)*, Association for Computational Linguistics, Vienna, Austria.

Siyu Liang, Gina-Anne Levow. 2025. Tone in Perspective: A Computational Typological Analysis of Tone Function in ASR. In *the 7th Workshop on Research in Computational Linguistic Typology and Multilingual NLP (SIGTYP)*, Association for Computational Linguistics, Vienna, Austria.

Siyu Liang, Nicolas Ballier, Gina-Anne Levow, Richard Wright. 2025. Beyond WER: Probing Whisper's Sub-token Decoder Across Diverse Language Resource Levels. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, Suzhou, China.

Siyu Liang. 2021. Documenting Eynu: A Case Study of Language Contact. In *Proceedings of the 43rd Annual Penn Linguistics Conference.*

Other Presentations

Siyu Liang, Zhaxi Zerong. 2025. De-toning Tibetan Dialects: A Computational Perspective on Tibetan Tonogenesis. In *the 58th International Conference on Sino-Tibetan Languages and Linguistics (ICSTLL 58)*, Bern, Switzerland. (Oral presentation)

Siyu Liang. 2025. Acoustic correlates of performed sexual orientation in Thai media. In *New Ways of Analysing Variation Asia Pacific (NWAV-AP 8)*, Singapore. (Oral presentation)

Siyu Liang. 2023. A Phonological Account of Intervocalic Voicing in Yangon Burmese. In *56th International Conference on Sino-Tibetan Languages and Linguistics (ICSTLL 56).* (Oral presentation)

Siyu Liang. 2021. Incomplete Vowel Assimilation in Tehrani Persian. In *4th Conference of Central Asian Language and Linguistics (ConCALL-4).* (Oral presentation)

Other Projects

September 2024 - Present

SocioLinx:
PI: Dr. Alicia Beckford Wassink (University of Washington)
A large-scale sociophonetic analysis of ASR errors for speech by underrepresented regional and ethnic groups of English speakers in the US. Benchmarking ASR models based on proprietary dataset.

September 2023 - Present

TurkicNLP at UW
PI: Dr. Talant Mawkanuli (University of Washington)
Building multilingual speaking dictionaries for Turkic languages. Providing mentorship on data annotation, NLP pipeline, and digital humanities practices.

Teaching

Spring 2026 — LING 471: Computational Methods for Linguists, University of Washington (Instructor of Record, scheduled)

Winter 2024 — LING 572: Advanced Statistical Methods in Natural Language Processing, University of Washington (Teaching Assistant)

Autumn 2023 — LING 570: Shallow Processing Techniques for Natural Language Processing, University of Washington (Teaching Assistant)

Spring 2023 — LING 471: Computational Methods for Linguists, University of Washington (Teaching Assistant, guest lecture on Statistics Basics for NLP)

Spring 2022
Autumn 2021 — LING 200: Introduction to Linguistics, University of Washington (Teaching Assistant, with teaching duties twice weekly)

Industry experience

Summer 2020 — **ByteDance (NLP Engineer Intern)** – Beijing, China
Building a FrameNet-based intent framework for large-scale dialog systems. Developing an extendable symbol set for multilingual NLP tasks.

Honors and scholarships

2024 — Imre Boba Summer Research Fellowship ($1000, University of Washington)

2021 — ConCALL-4 Travel Award ($500)

2019 — Royden B. Davis Fellowship ($5000, Georgetown University)

2018 — Lisa J. Raines Fellowship ($5000, Georgetown University)

Talks and tutorials

2025 — Analyzing Whisper Across Diverse Language Resource Levels
Workshop on Promoting Fairness for Under-represented Languages in Multilingual LLMs, May 26 2025, Université Paris Cité and Online

2025 — Languages of the Silk Road (Guest lecture)
MELC 268: Silk Road and Globalization, Spring 2025, University of Washington

2024 Tibetan Music (Guest lecture)
MELC 269: Music Cultures of the Silk Road, Winter 2024, University of Washington

Mentorship

2024 – 2025 Ezra Mason, Linguistics Undergraduate Research Apprenticeship (LURAP) on *Automatic Segmentation of Tuvan*

2024 – 2025 Melody Vahadi, M.S. project on *OCR for Chaghatay*

2024 – 2025 Benjamin Pong, M.S. project on *Continued Pre-training for Low Resource Language ASR*

Service

2025 Editor, *Proceedings of the 43rd West Coast Conference on Formal Linguistics*

2023 – 2024 Treasurer, *Linguistic Society of the University of Washington (LSUW)*

Technical skills

Programming languages
Proficient in: Python.
Familiar with: C++, Java, JavaScript, R.

Software
ELAN, Praat

Natural Languages
Proficient: Mandarin, English
Advanced: Persian, French (reading)
Intermediate: Thai, Tibetan, Vietnamese, Indonesian, Japanese
Elementary: Burmese, Uyghur, Uzbek, Arabic, Amharic, Turkish, Russian, Portuguese, Swahili
Experience: Nobiin, Chaghatay, Tuvan, Nepali, Mongolian, etc.